SMART CAMPUS EVENT MANAGEMENT SYSTEM

School of Computing

Bachelor of Technology
in
Computer Science & Engineering

By

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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled ”SMART CAMPUS EVENT MANAGEMENT SYSTEM” by SUPRAJA K (VTU25927) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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DECLARATION

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ABSTRACT

The Smart Campus Event Management System is a web-based application de-signed to streamline the creation, management, and registration of campus events in universities and colleges. It provides a centralized platform where students can securely register, explore upcoming events, apply advanced search and filtering op-tions, and manage their event participation. Administrators are provided with ded-icated tools to create, update, and delete events, monitor registrations, and analyze participation through real-time statistics. The system incorporates modern technolo-gies such as a React-based frontend and a Spring Boot backend with RESTful APIs, ensuring efficient communication and scalability. By replacing manual event man-agement processes, the system minimizes errors, avoids overbooking through ca-pacity constraints, saves time, and enhances overall user experience and operational efficiency.

Keywords: Smart Campus, Event Management System, React.js, Spring Boot, REST API, User Authentication, Event Registration, H2 Database, Axios, Single Page Ap-plication, CRUD Operations, Web Application, Automation

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
Auth	Authentication
CRUD	Create, Read, Update, Delete
DB	Database
HTTP	HyperText Transfer Protocol
JPA	Java Persistence API
JSON	JavaScript Object Notation
MVC	Model View Controller
REST	Representational State Transfer
UI	User Interface

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Chapter 1

INTRODUCTION

1.1 Introduction

In the era of digital transformation, educational institutions are increasingly adopt-ing smart solutions to streamline administrative and academic activities. One of the key areas that still relies on semi-manual processes is campus event management, which includes event creation, registration, and participation tracking. Traditional methods such as manual registrations, spreadsheets, and notice boards are time-consuming, error-prone, and inefficient when handling large numbers of students.

The Smart Campus Event Management System is developed as a modern web-based solution to overcome these challenges. It provides a centralized platform where students can explore, filter, and register for events, while administrators can efficiently manage event creation, monitor registrations, and analyze participation data.

The system leverages a full-stack architecture using modern technologies, ensur-ing scalability, responsiveness, and ease of use. It enhances transparency, reduces manual workload, prevents issues like overbooking through capacity constraints, and improves overall user experience for both students and administrators.

1.2 Aim of the Project

The primary aim of the Smart Campus Event Management System is to design and develop a user-friendly and efficient web-based platform for managing campus events and student participation. The project focuses on achieving the following objectives:

- To automate the event management process within educational institutions
- To provide a centralized platform for students to explore and register for events
- To enable administrators to efficiently create, update, and manage events
- To reduce manual errors and prevent overbooking using capacity constraints
- To improve overall efficiency, accessibility, and user experience

1.3 Scope of the Project

The Smart Campus Event Management System focuses on developing a web-based platform that enables efficient management of campus events and student participation within educational institutions. The system allows students to explore available events, apply advanced search and filtering based on criteria such as event name, date, and department, and register for events online while ensuring capacity constraints are maintained to prevent overbooking. It also provides administrators with tools to create, update, and delete events, as well as monitor all student regis-trations and analyze participation through basic statistics. The platform is designed using a modern full-stack architecture with a React-based frontend, a Spring Boot backend, and an H2 database for data storage. The system aims to reduce manual effort, improve accuracy, and streamline event coordination, while being scalable for future enhancements such as notifications, analytics, and integration with other institutional systems.

1.4 Framework

The system framework consists of multiple layers including the user interface, application logic, and database. The frontend layer is developed using React.js to provide an interactive user experience. The backend layer is built using Spring Boot, which handles business logic and RESTful API communication. The database layer uses an H2 in-memory database to store event, student, and registration data efficiently.

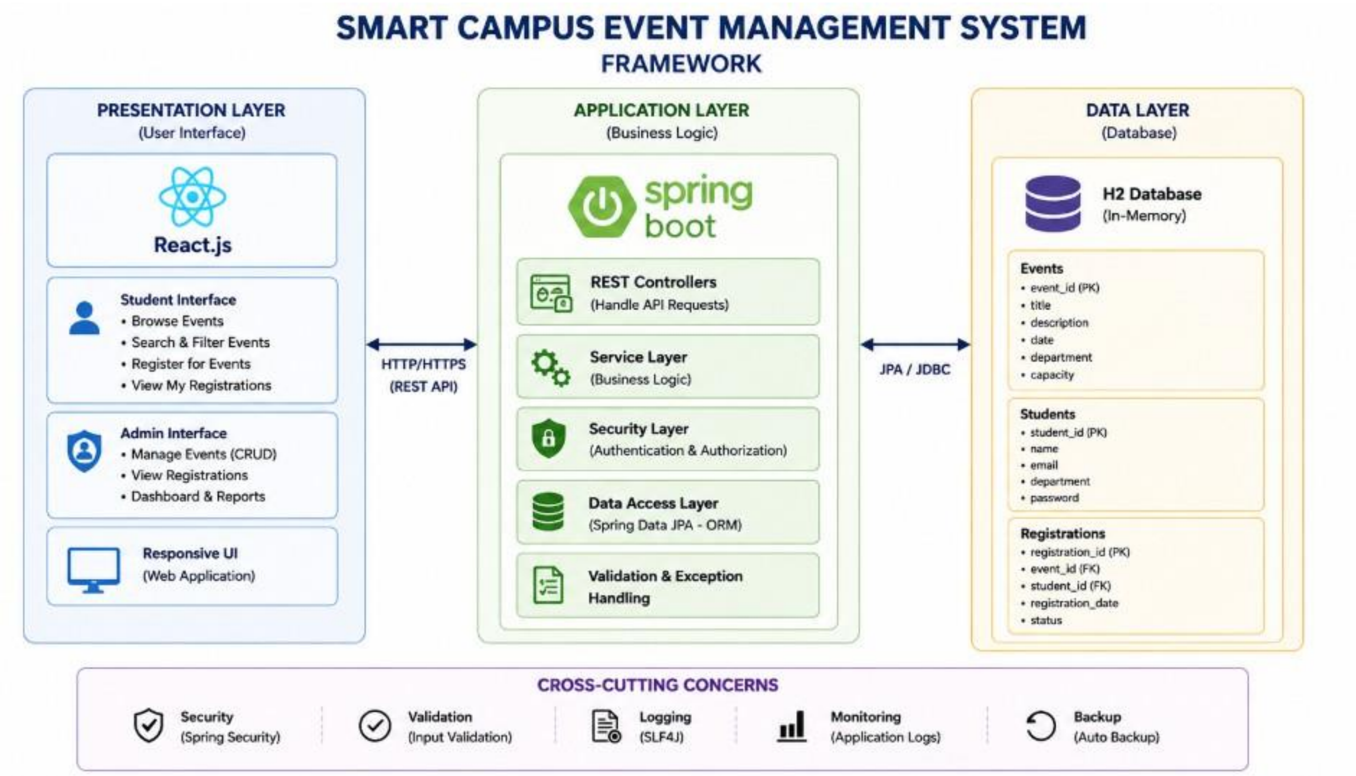

Figure 1.1: Framework

Figure 1.1 says that the Smart Campus Event Management System showcasing presentation, application, and data layers with integrated REST APIs and security features. It highlights user interaction through React.js, backend processing via Spring Boot, and data management using the H2 database with cross-cutting con-cerns like logging, validation, and monitoring.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

Several event management and registration applications are widely used in the market, such as Eventbrite, Meetup, Townscript, and Cvent. These platforms provide essential features including event creation, online registration, ticket management, and user-friendly interfaces that enhance overall user experience.

In addition to these, platforms like Whova and Bizzabo offer advanced functionalities such as attendee engagement tools, event analytics, networking features, and automated communication. These features help organizers efficiently manage large-scale events and track participant interaction.

Furthermore, institutional portals and basic registration systems are also used within universities for managing internal events. These systems provide simple functionalities such as event listing, manual registration, and limited tracking mecha-nisms. However, they often lack scalability, modern interfaces, and advanced filter-ing capabilities.

Most existing applications are primarily designed for large-scale public events such as conferences, exhibitions, and commercial gatherings. They commonly in-clude features like ticket booking, payment integration, analytics dashboards, and promotional tools that support large user bases.

However, these platforms often lack specific support for academic or campus-level event management, where requirements such as department-based filtering, re-stricted access, capacity management, and simplified workflows are essential. This gap highlights the need for a dedicated system tailored for campus environments.

Feature	**BookMyShow**	**Eventbrite**	**Townscript**	**Your Project**
Event Creation	Yes	Yes	Yes	Yes
Ticket Booking	Yes	Yes	Yes	Yes
Internal Department Events	No	No	No	Yes
User Registration/Login	Yes	Yes	Yes	Yes
Online Payment Integration	Yes	Yes	Yes	No / Planned
QR Code / E-Ticket	Yes	Yes	Yes	Yes
Seat Selection	Yes	Limited	Limited	Yes
Event Analytics	Yes	Yes	Yes	Basic
Direct Communication (Admin-User)	No	Limited	No	Yes

Table 2.1: Comparative Analysis of Market Alternatives

The above Table 2.1 compares key features of existing platforms with the proposed system, showing that most basic functionalities are supported across all.However, the project stands out by offering internal department events, better seat selection, and direct admin-user communication tailored for campus needs.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend development environment is built using modern web technologies such as React.js and Vite. Node.js is used as the runtime environment for managing dependencies and running the development server. Tools like Visual Studio Code are used for coding, while web browsers are used for testing and debugging. Additional libraries such as Axios are used for API communication, and React Router DOM is used for navigation within the application.

3.1.2 Structure of the Project

The frontend follows a modular and component-based architecture to ensure scalability, maintainability, and efficient development.

The public directory contains static files such as the main HTML file and images used across the application. The source directory is organized into multiple folders, including components, pages, and services.

Reusable UI components such as navigation bars, event cards, and forms are stored in the components folder. The pages folder contains the main screens of the application, including the landing page, student dashboard, admin dashboard, login, and registration pages. The services folder manages API calls using Axios, ensuring smooth communication with the backend server.

This structured approach helps in maintaining clean code and enables easy expansion of features.

3.1.3 Execution Procedure

The frontend application is executed by running the development server using Node.js. Once the server starts, the application is accessible through a web browser. Users can interact with the interface to browse events, apply filters, register for events, and access dashboards. Developers can test and debug features in real time during execution.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Spring Boot, which provides a robust frame-work for building RESTful APIs and handling business logic. Java Development Kit (JDK) is required for development, and Maven is used for dependency manage-ment and project build lifecycle.

Spring Boot simplifies configuration and deployment through its embedded server. Tools such as Postman are used for testing API endpoints, and Git is used for version control and collaboration.

3.2.2 Structure of the Project

The backend project follows a layered architecture consisting of controllers, services, repositories, and model classes.

The controller layer handles incoming HTTP requests and defines API endpoints. The service layer contains business logic such as event management, registration handling, and validation. The repository layer uses Spring Data JPA to interact with the database without writing complex SQL queries. The model layer defines entity classes such as Student, Event, and Registration.

This structured approach ensures separation of concerns, improves maintainabil-ity, and supports scalability.

3.2.3 Execution Procedure

The backend server is started using Maven commands or directly from an IDE. Once the server is running, it listens for client requests and processes them through defined REST APIs. The backend communicates with the database to store and retrieve data, and responses are sent back to the frontend in JSON format

3.3 Database Implementation

3.3.1 Database Configuration

The system uses an H2 in-memory database for storing application data. It is configured within the Spring Boot application using properties files. The database automatically initializes during runtime and provides fast data access without requiring external setup.

3.3.2 Schema Structure

The database schema is designed using a relational model, organizing data into structured tables with proper relationships. Entities such as students, events, and registrations are connected using primary and foreign keys to maintain data integrity and consistency.

3.3.3 Tables created

The core functionalities are supported by the following tables:

- **Student Table:** Stores student details such as ID, name, email, department, and password for authentication.
- **Event Table:** Contains event information including title, date, department, type, description, and capacity.
- **Registration Table:** Acts as a link between students and events, storing details such as registration ID, student information, and event reference.

3.4 System Technology Stack Overview

The following table provides a comprehensive summary of the frameworks and technologies utilized in the development of the Smart Campus Event Management System, categorized by their architectural role.

Component	**Technology**
Frontend	React.js, Vite, HTML5, CSS3, React Router DOM
Backend	Spring Boot, Spring Security, Spring Data JPA
Database	H2 In-Memory Database
API Communication	Axios
Tools	Git, VS Code

Table 3.1: System Technology Stack

The following table 3.1 provides a comprehensive summary of the frameworks and technologies utilized in the development of the Smart Campus Event Management System, categorized by their architectural role.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Smart Campus Event Management System follows a layered architecture consisting of frontend, backend, and database components. The frontend is responsible for handling user interactions and providing a dynamic interface for students and administrators. The backend manages business logic, processes requests, and handles communication between the frontend and the database. The database stores all essential data such as student details, event information, and registrations. This structured architecture ensures efficient data flow, improved security, and scalability of the system. Additionally, RESTful APIs enable seamless communication between different layers, ensuring real-time data exchange. Security mechanisms such as authentica-tion and authorization protect sensitive information and restrict unauthorized access. The modular design of the system allows easy maintenance, future enhancements, and integration of new features.

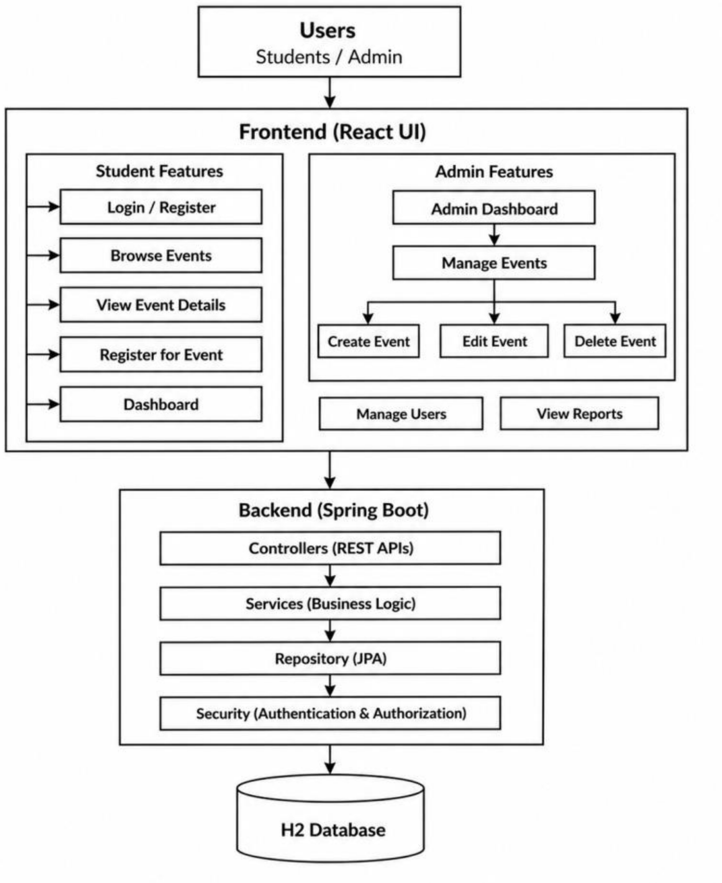

Figure 4.1: System Architecture

Figure 4.1 says that the Layered architecture separating presentation (React.js UI), application (Spring Boot REST services), and data (H2 database) for efficient event management.Ensures secure communication, scalable design, and seamless handling of user interactions, business logic, and data storage.

4.2 DataFlow Diagram

The Data Flow Diagram illustrates how data moves within the system. User in-puts, such as login details or event registration requests, are sent from the frontend to the backend through API calls. The backend processes these requests, performs nec-essary validations such as checking event capacity, and interacts with the database to store or retrieve information. The processed data is then sent back to the frontend, which displays outputs such as event lists, registration confirmations, and user dashboards.

4.2.1 DFD Level 0: Context Diagram

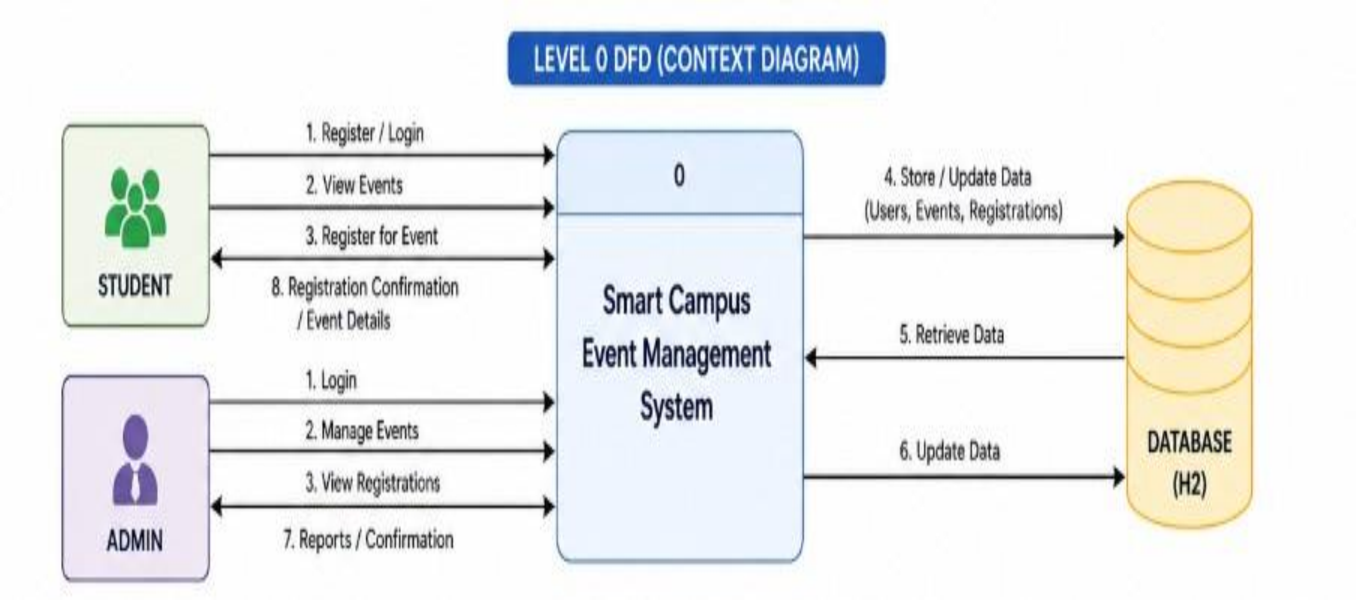

Figure 4.2: Data Flow Diagram Level 0: Context Diagram

Figure 4.2 says that the Level 0 Data Flow Diagram represents the overall system as a single process interacting with external entities such as students and adminis-trators. It shows how user inputs like login, event requests, and registrations are processed by the system and how responses are returned through the database.

4.2.2 DFD Level 1: Functional Decomposition

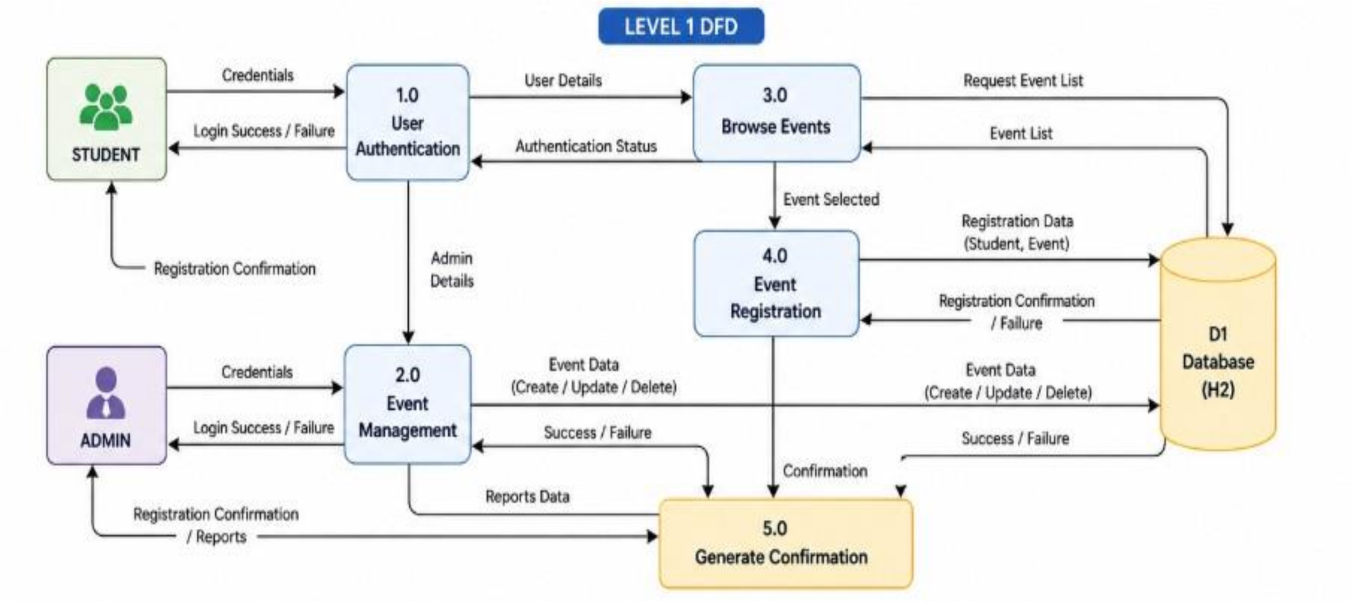

Figure 4.3: Data Flow Diagram Level 1: Functional Decomposition

Figure 4.3 says that the Level 1 Data Flow Diagram provides a detailed breakdown of the internal processes of the system, including user authentication, event brows-ing, event management, and event registration. It illustrates how data flows between these processes and the database, ensuring proper validation, storage, and retrieval of information for efficient system operation.

4.3 System Modules

The system is divided into key modules including User Management, Event Management, and Event Registration to ensure organized functionality. Each module handles specific operations such as authentication, event creation, and student registrations. This modular approach improves system efficiency, scalability, and ease of maintenance.

4.3.1 Module 1: User Management

This module handles student registration, login, and authentication processes. It ensures that only authorized users can access the system and perform actions such as event registration. The module securely stores user credentials and validates login information. It also manages user sessions and maintains data integrity.

4.3.2 Module 2: Event Management

This module is designed for administrators to manage events efficiently. It allows admins to create new events, update existing event details, and delete events when necessary. Information such as event title, date, department, type, description, and capacity is maintained. This module ensures proper organization and control over all campus activities.

4.3.3 Module 3: Event Registration

This module enables students to register for events easily. It allows users to view event details, check seat availability, and complete registration. The system validates each registration to prevent duplicate entries and ensures that the event capacity is not exceeded. Once registration is successful, confirmation details are provided to the user.

4.4 Advantages and Limitations of the Project

The Smart Campus Event Management System offers several advantages by automating and simplifying the event management process. It reduces manual work-load, improves efficiency, and ensures accurate data handling. The system provides a user-friendly interface for both students and administrators, making event coordination seamless.

4.4.1 Advantages

- **Time Saving:** Automates event management and registration processes, reducing manual effort.

- **Error Reduction:** Minimizes human errors and prevents issues such as duplicate registrations and overbooking.

- **Easy Management:** Enables administrators to efficiently manage events and track student participation.

4.4.2 Limitations

The system depends on internet connectivity for proper functioning, and performance may be affected in case of network issues. Additionally, initial development and setup require technical knowledge and time. Regular maintenance is necessary to ensure smooth operation and to handle potential system issues.

Chapter 5

RESULTS AND DISCUSSIONS

5.1 System Implementation Overview

The Smart Campus Event Management System is designed as a comprehensive web-based platform integrating multiple modules for both students and administra-tors to efficiently manage campus events and participation.

- Landing Page Interface
- Authentication Module
- Student Dashboard
- Admin Dashboard
- Event Management Module
- Event Registration Module
- Advanced Search & Filtering
- My Registrations (Student Panel)
- Analytics Dashboard
- Notification System
- Database Implementation

5.2 System Interface Visualizations

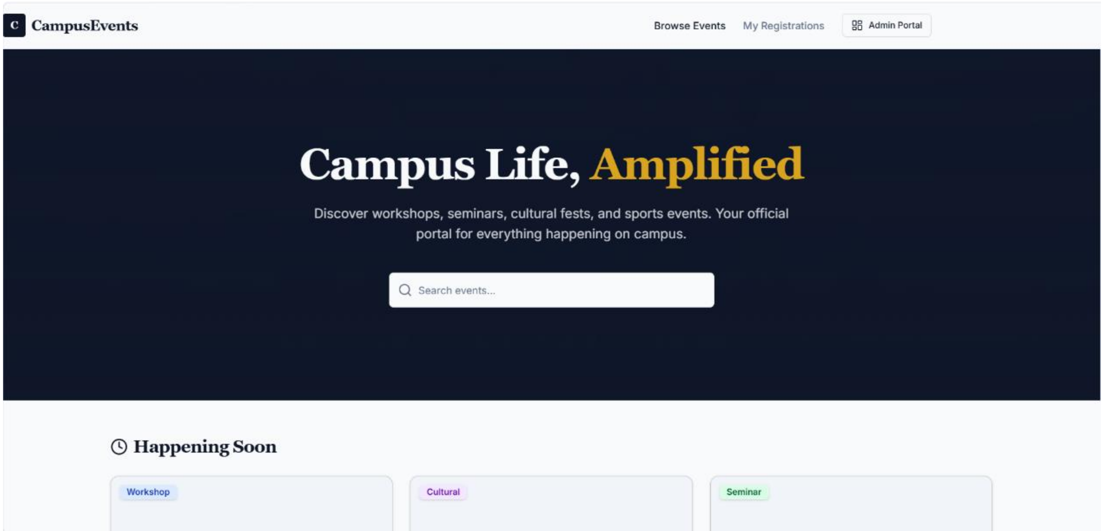

Figure 5.1: Landing Page Interface

Figure 5.1 says that to Discover and register for campus events like workshops, seminars, and networking sessions with ease.

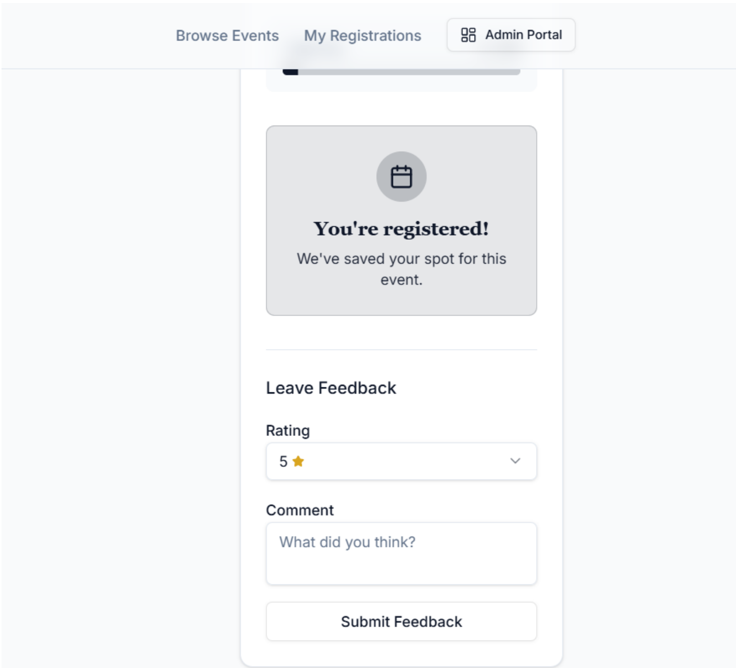

Figure 5.2: Login Interface

Figure 5.2 says that the login page allows users to securely access the system using their credentials with proper authentication.

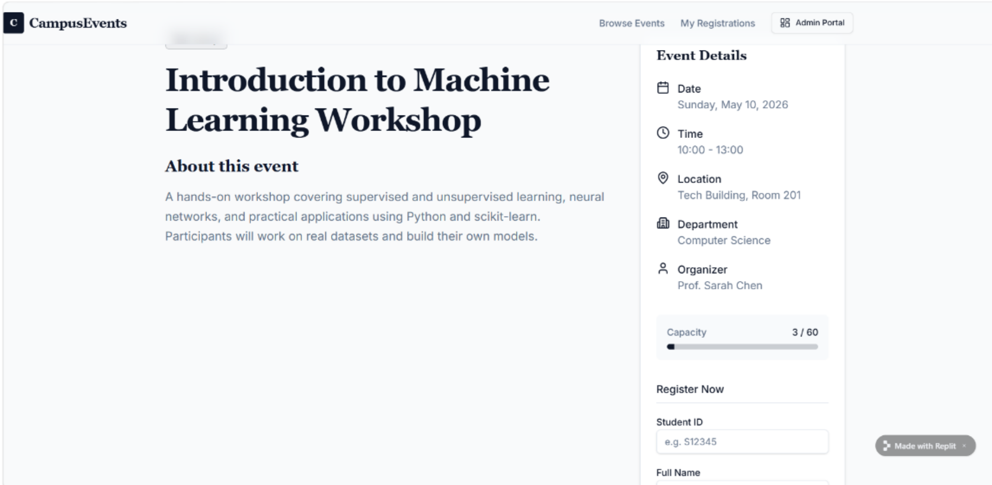

Figure 5.3: Signup Interface

Figure 5.3 says that the registration page enables new users to create an account by entering required personal details.It ensures data accuracy through input validation and stores user information securely in the database.

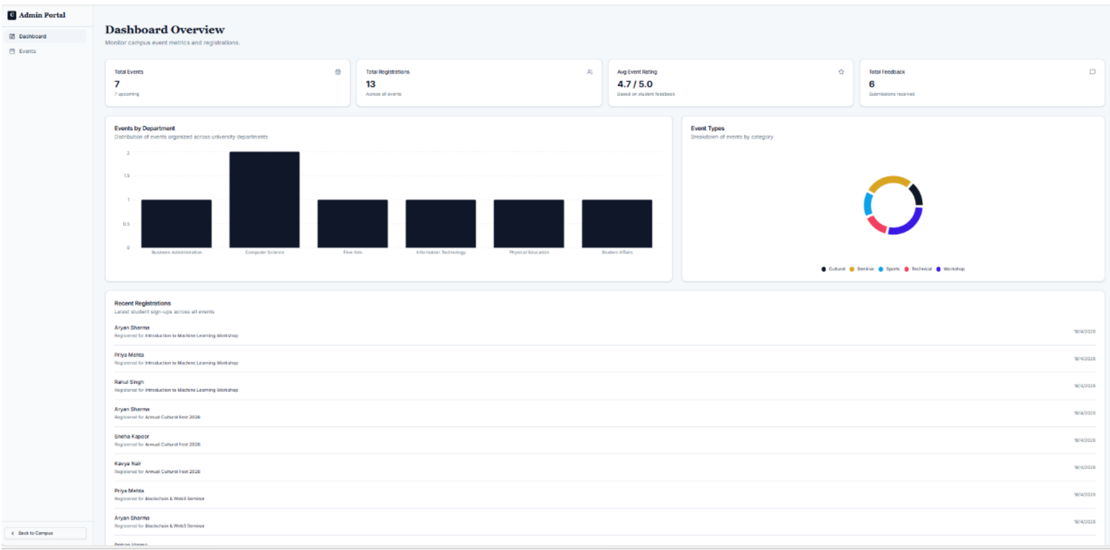

Figure 5.4: Admin Dashboard

Figure 5.4 says that the admin dashboard provides a centralized interface to manage events, users, and registrations efficiently.

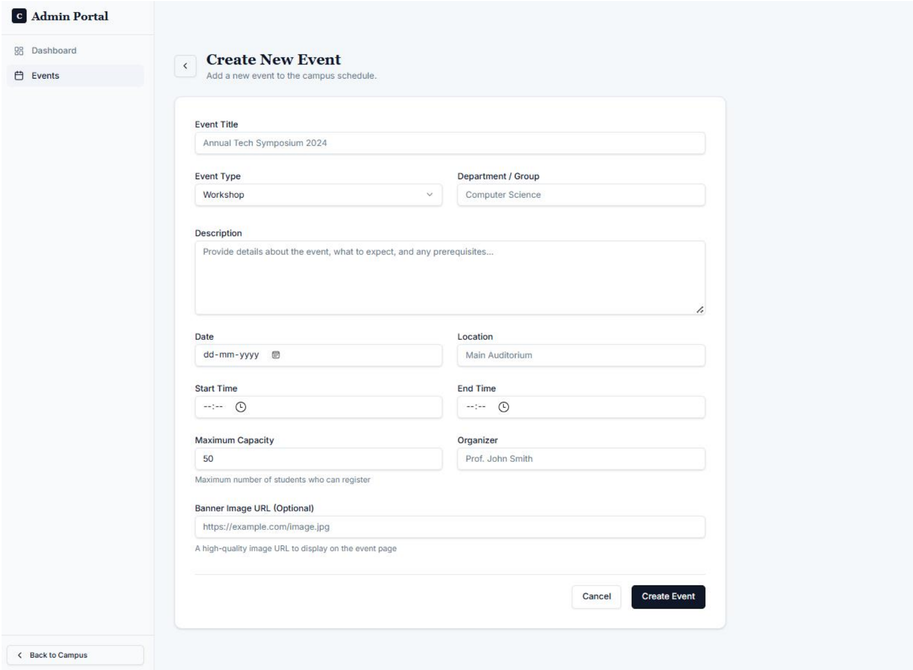

Figure 5.5: Events Creation Page

Figure 5.5 says that the event creation page allows administrators to add new events by entering details such as title, date, venue, and description.

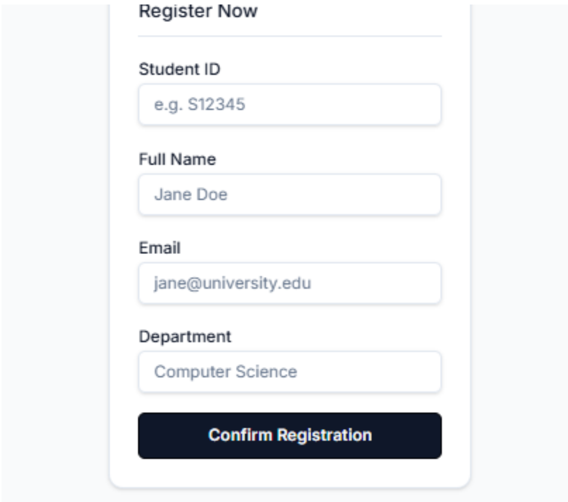

Figure 5.6: View Registration Page

Figure 5.6 says that this page displays the list of students who have registered for a specific event with details like name, ID, and contact information. It enables administrators to view, manage, and track registrations efficiently, with options to filter or export data if needed.

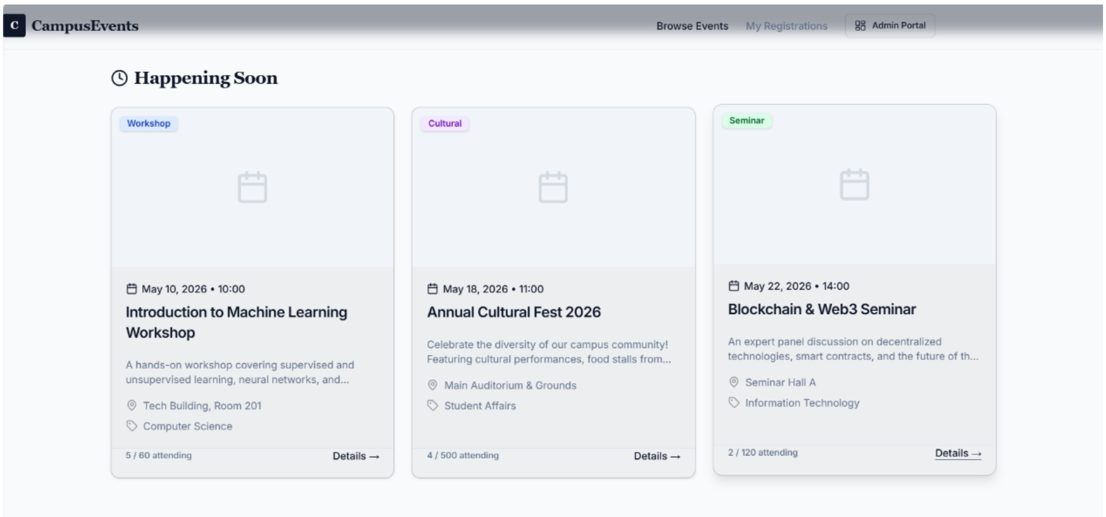

Figure 5.7: Student Dashboard Page

Figure 5.7 says that the student dashboard provides a personalized interface to view available events, registered events, and important updates.

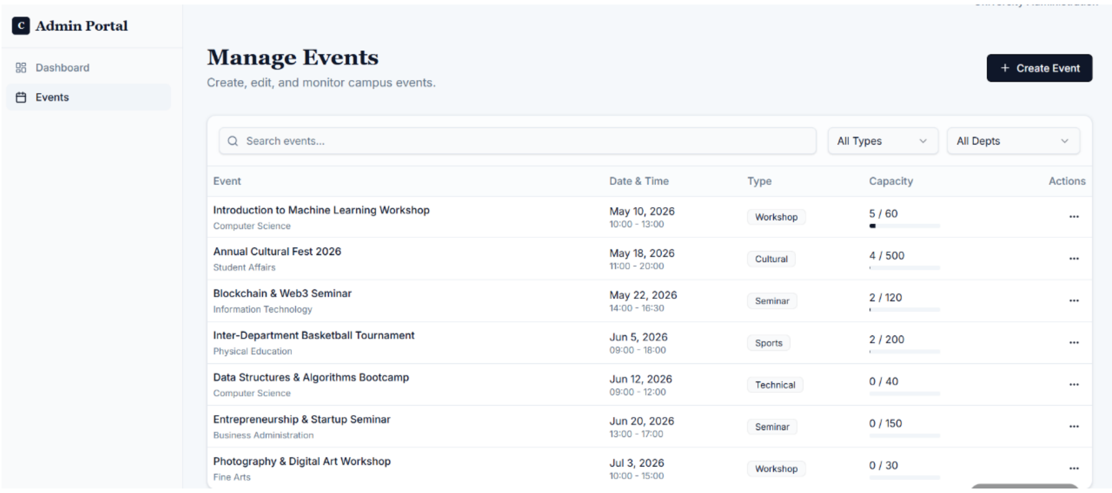

Figure 5.8: Student Registration Page

Figure 5.8 says that the student registration page allows users to sign up for events by providing necessary details such as name and mail information. It ensures accu-rate submission through validation and confirms successful registration while updat-ing the database in real time.

5.3 Discussion

The implementation of the Smart Campus Event Management System demon-strates a significant improvement over traditional event management methods used in educational institutions. This section discusses the integration of various modules, system performance, and overall impact on event coordination.

5.3.1 Efficiency and Automation

The system successfully automates the entire event management process, includ-ing event creation, registration, and participation tracking. By replacing manual methods, it reduces paperwork, minimizes human errors, and saves time for both students and administrators. The automation ensures smooth handling of multiple events simultaneously without data inconsistencies.

5.3.2 User Experience and Accessibility

The application provides a user-friendly and responsive interface, allowing stu-dents to easily browse events, apply filters, and register with minimal effort. The admin dashboard simplifies event management tasks, making it easier to monitor registrations and update event details. The use of a Single Page Application enhances performance and provides a seamless user experience.

5.3.3 System Performance and Data Handling

The system ensures efficient data handling through structured database design and backend processing. The use of REST APIs enables smooth communication between frontend and backend, while validation mechanisms prevent issues such as duplicate registrations and overbooking. The system performs reliably under normal usage conditions.

5.3.4 Architectural Design and Scalability

The layered architecture of the system allows clear separation of concerns between frontend, backend, and database layers. This design improves maintainability and scalability, making it easier to use.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Smart Campus Event Management System successfully achieves its objective of providing a centralized and efficient platform for managing campus events and student participation. The system replaces traditional manual processes with a digital solution that simplifies event creation, registration, and monitoring.

The implementation demonstrates that using a full-stack architecture with a React-based frontend and a Spring Boot backend ensures smooth communication, efficient data handling, and a responsive user experience. Features such as authentication, event management, advanced filtering, and registration validation contribute to a re-liable and user-friendly system.

Overall, the project improves operational efficiency, reduces manual errors, and enhances coordination between students and administrators. It serves as a scalable and practical solution for educational institutions to manage events effectively.

6.2 Future Enhancements

While the current implementation of the Smart Campus Event Management Sys-tem provides a strong foundation for managing campus events and student registra-tions, the following enhancements are planned for future versions:

- **Real-Time Notification System:** Implementation of real-time notifications to inform users about event updates, reminders, and registration status using technologies like WebSockets.
- **Payment Integration:** Integration of secure payment gateways to support paid events and enable seamless online transactions.
- **QR Code-Based Attendance System:** Enhancing event participation tracking by implementing QR code-based check-in for faster and more accurate atten-dance management.
- **Advanced Analytics Dashboard:** Introducing detailed analytics for administrators, including participation trends, popular events, and department-wise insights for better decision-making.
- **Email and SMS Notification System:** Automated alerts for event confirmations, reminders, cancellations, and updates to improve communication and user engagement.
- **Role-Based Access Control Enhancement:** Expanding user roles such as Su-per Admin and Event Organizer to improve system security and provide better control over operations.
- **Mobile Application Development:** Developing a mobile application to im-prove accessibility and provide a seamless user experience across devices.
- **Scalable Database Integration:** Migrating from H2 in-memory database to scalable databases such as MySQL or PostgreSQL for handling large-scale data efficiently.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

The Smart Campus Event Management System addresses a complex engineering problem by integrating frontend, backend, and database technologies to automate campus event management. It replaces manual processes with a scalable digital solution while handling constraints such as user authentication, event capacity, and real-time data processing. The system supports multiple stakeholders including stu-dents and administrators, ensuring efficient and reliable operation.

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of knowl-edge	In-depth engineer-ing knowledge	WK3–WK6	Uses React.js, Spring Boot, REST APIs, and H2 database for full-stack implementation.
WP2	Conflicting re-quirements	Technical vs non-technical conflicts	WK4–WK6	Balances user-friendly interface with constraints like event capacity and secure authentication.
WP3	Depth of Analy-sis	Analytical and de-sign thinking	WK3–WK5	Designs relational schema linking students, events, and registrations with validation checks.
WP4	Familiarity	Partially new prob-lems	WK4, WK8	Converts manual event handling into an automated web-based sys-tem with dashboards.
WP5	Applicable Codes	Limited standar d solutions	WK6	Implements filtering, CRUD opera-tions, and secure API-based work-flows.
WP6	Stakeholde r Needs	Multiple stakehold-ers	WK5–WK7	Supports students (registration) and admins (event management and an-alytics).
WP7	Interdependence	System integration	WK3, WK5, WK6	Integrates frontend, backend, and database for seamless real-time communication.

Table 7.1: Mapping of Smart Campus Event Management System to CEP

The above table 7.1 maps the Smart Campus Event Management System to Complex Engineering Problem (CEP) attributes, highlighting its alignment with various knowledge, analysis, and design requirements. It demonstrates how the project integrates full-stack technologies, stakeholder needs, and system-level thinking to address real-world engineering challenges effectively.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes React.js, Vite, Spring Boot, Spring Data JPA, and RESTful APIs.
EA2	Level of Interactions	Complex interactions between modules	Manages event lifecycle including creation, registration, and capacity validation.
EA3	Innovation	Application of modern technologies	Replaces manual systems with a Single Page Application (SPA) and automated workflows.
EA4	Consequences to So-ciety and Environment	Impact on users and society	Reduces manual work, improves efficiency, and minimizes paper usage.
EA5	Familiarity	Beyond standard practices	Implements scalable architecture capable of handling multiple users and events.

Table 7.2: Mapping of Smart Campus Event Management System to CEA

Table 7.2 says that the project involves complex engineering activities such as system design, integration of multiple technologies, and implementation of scalable architecture. It demonstrates innovation by transforming traditional event management into a digital platform with real-time updates and efficient workflows.

7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Encourages student participation in academic and extracurricular events, enhancing learning experience.
SDG 9	Industry, Innovation and Infrastructure	Implements modern web technologies to digitize campus event management.
SDG 11	Sustainable Cities and Communities	Supports a smart campus by providing an efficient and paperless system.
SDG 12	Responsible Consumption and Production	Reduces paper usage by digitizing event registrations and records.
SDG 17	Partnerships for the Goals	Enhances collaboration between students and administrators through a centralized platform.

Table 7.3: SDG Mapping for Smart Campus Event Management System

Table 7.3 says that the Smart Campus Event Management System contributes to sustainable development by promoting digital transformation, improving accessibil-ity to events, and reducing paper-based processes within educational institutions.

Chapter 8

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